2023 BECE MATHEMATICS PAPER ONE

OBJECTIVES

1. Write 0.01723 in the standard form.
   A. 0.01723×10^{-2}
   B. 0.01723×10^{2}
   C. 1.723×10^{-2}
   D. 1.723×10^{2}
2. What is the value of $3x^2 + 2x - 7$ when $x = -3$?
   A. – 40
   B. – 33
   C. 14
   D. 26
3. Find the image of the point S(-2, 2) when it is reflected in the y-axis.
   A. (2, -2)
   B. (2, 2)
   C. (-2, -2)
   D. (-2, 2)
4. What is the actual bearing of S 25^0 E?
   A. 035^0
   B. 065^0
   C. 115^0
   D. 155^0
5. Given that $m=\frac{b}{2} - t$, make t the subject.
   A. $T = \frac{b-2m}{2}$
   B. $T = \frac{2m-b}{2}$
   C. $T = \frac{2b-m}{2}$
   D. $T = \frac{b+2m}{2}$
6. Expand $(x - 2)(x + 4)$
   A. $\chi^2 + 2\chi - 8$
   B. $\chi^2 + 2\chi + 8$
   C. $\chi^2 - 4\chi - 8$
   D. $\chi^2 - \chi - 8$
7. If n(E) = 15, n(F) = 20 and n(E ∩ F) = 6, find n(E U F).
   A. 21
   B. 26
   C. 29
   D. 35
8. Tony shared $\frac{2}{3}$ of his plot of land equally among his three sons. What fraction of the plot did each get?
   A. $\frac{2}{9}$
   B. $\frac{1}{3}$
   C. $\frac{1}{2}$
   D. $\frac{5}{9}$
9. Araba and Ama share 20 oranges in such a way that Ama receives 12 oranges. Find the ratio of Araba's share to that of Ama.
   A. 2 : 5
   B. 3 : 5
   C. 3 : 2
   D. 2 : 3
10. Solve: $2(\chi - 2) - \frac{\chi}{3} + 6 > 0$
    A. $X < 0$
    B. $X > 0$
    C. $X > 3$
    D. $X > \frac{1}{8}$

The stem and leaf plot shows the marks scored by students in a French test. Use the information to answer questions 11 to 13

Stem	Leaf
2	0 2 5 7 8
3	2 7 9
4	3 5 5 5
5	4 6 6 8
6	3 5 7
7	0 6

11. What is the model mark?
    A. 35
    B. 45
    C. 56
    D. 76
12. How many students took the text?
    A. 20
    B. 21
    C. 22
    D. 27
13. Find the median mark.
    A. 45
    B. 47
    C. 54
    D. 76
14. Which of the following is an **even** prime number?
    A. 2
    B. 1
    C. 4
    D. 8
15. If $2n - 5 = \frac{1}{2}n$, find the value of n.
    A. $\frac{1}{3}$

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B. $\frac{1}{2}$
C. $2\frac{1}{2}$
D. $3\frac{1}{3}$

16. Find the simple interest on GH₵ 350.00 for 4 years at 5% per annum.
A. GH₵ 20.00
B. GH₵ 35.00
C. GH₵ 70.00
D. GH₵ 140.00

17. A straight line passes through the points P(-5, -3) and Q(-4, -7). Find the gradient of the line. PQ
A. -4
B. $-\frac{1}{4}$
C. $\frac{1}{4}$
D. 4

18. Given that 1 kilometre = $\frac{5}{8}$ mile, what is 240 miles in kilometres?
A. 150km
B. 190km
C. 384km
D. 390km

19. John walks for $22\frac{1}{2}$ minutes and runs for $7\frac{1}{2}$ minutes to school. What percentage of the total time does he spend walking?
A. 25 %
B. 30 %
C. 33 %
D. 75 %

20. Kobby is 5years older than his brother. If Kobby is 13years old, how old is his brother?
A. 7 years
B. 8 years
C. 10 years
D. 18 years

21. Find the rule of the mapping:

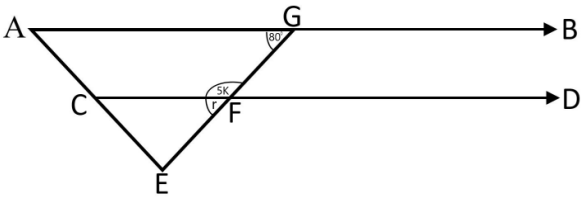
X	1	2	3	4	5
↓	↓	↓	↓	↓	↓
Y	4	2	0	-2	-4

A. Y = 2x + 2
B. Y = -2x + 2
C. Y = 4x
D. Y = -2x + 6

22. Arrange the following numbers in ascending order: $\frac{9}{5}$, 1.88, $\frac{15}{8}$
A. $\frac{9}{5}$, $\frac{15}{8}$, 1.88
B. $\frac{15}{8}$, 1.88, $\frac{9}{5}$
C. 1.88, $\frac{9}{5}$, $\frac{15}{8}$
D. $\frac{9}{5}$, 1.88, $\frac{15}{8}$

23. Simplify: $(x - 1)^2 - 1$.
A. $\chi^2 - 2\chi$
B. $\chi^2 + 2\chi$
C. $\chi^2 - 2\chi - 1$
D. $\chi^2 - 2\chi + 1$

24. A bag contains 20 oranges of which 6 are bad. Find the probability of picking a good oranges from the bag.
A. $\frac{1}{20}$
B. $\frac{3}{10}$
C. $\frac{1}{6}$
D. $\frac{7}{10}$



NOT DRAWN TO SCALE

In the diagram, AB is parallel CD. Angle AGE is 80^0, angle CFG = 5K and angle CFE = r.

Use the diagram to answer question **25** and **26.**

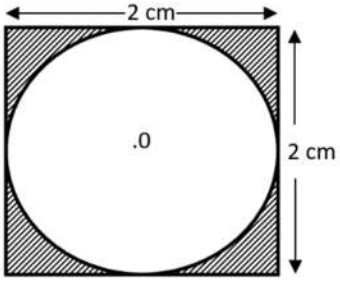
25. What is the value of r?
A. 16^0
B. 20^0
C. 80^0
D. 160^0

26. Find the value of k.
A. 16^0
B. 20^0
C. 80^0
D. 160^0

27. Simplify: $(1\frac{1}{2} - \frac{5}{6}) \times \frac{9}{10}$.
A. $\frac{1}{10}$
B. $\frac{3}{20}$
C. $\frac{1}{5}$
D. $\frac{3}{5}$

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28. Factorize $2pq + 6p - 6q - 18$.
    A. $2(p - 3)(q - 3)$
    B. $2(p + 3)(q + 3)$
    C. $2(p - 3)(q + 3)$
    D. $2(p + 3)(q - 3)$



NOT DRAWN TO SCALE

The diagram is a square of size 2cm in which is inscribed a circle with centre O. use the information to answer question 29 and 30.

29. Find the area of the shaded portion.
    A. 0.86 cm^2
    B. 3.00 cm^2
    C. 6.28 cm^2
    D. 12.56 cm^2
30. Find the area of the circle. [Take $\pi=\frac{22}{7}$]
    A. 3.14 cm^2
    B. 6.28 cm^2
    C. 4.00 cm^2
    D. 12.56 cm^2
31. A man take 12 hours to do a piece of work. How long will it take 6 men working at the same rate to do the work?
    A. 6 hours
    B. 3 hours
    C. 2 hours
    D. 72 hours
32. Convert 12.03 metres to centimetres.
    A. 0.1203 cm
    B. 120.3 cm
    C. 1203 cm
    D. 12030 cm
33. A quadrilateral with one pair of opposite sides parallel is called a
    A. Kite
    B. Rectangle
    C. Square
    D. Trapezium
34. A bool was sold for GH₵ 48.00 at a profit of 20%. Find the cost price.
    A. GH₵ 38.40
    B. GH₵ 40.00
    C. GH₵ 60.00
    D. GH₵ 57.60
35. Find the vector which translate the point (2, 6) to (5, 4).
    A. $\begin{matrix}-3\\2\end{matrix}$
    B. $\begin{matrix}-1\\4\end{matrix}$
    C. $\begin{matrix}3\\-2\end{matrix}$
    D. $\begin{matrix}7\\10\end{matrix}$
36. Cement and sand were mixed in the ratio 2 : 5. How many kilograms of cement was contained in the 35 kg of the mixture?
    A. 7kg
    B. 10kg
    C. 14kg
    D. 88kg
37. Which of the following is not an integer?
    A. 0
    B. 1
    C. 0.5
    D. -5
38. Simplify: $\frac{5^7 \times 5^{-4}}{5^3}$
    A. 0
    B. 1
    C. 5
    D. 25
39. The height of a cylinder is 5cm and the radius is 7cm. Find the volume of the cylinder. [Take $\pi=\frac{22}{7}$]
    A. 770 cm^2
    B. 700 cm^2
    C. 154 cm^2
    D. 110 cm^2
40. If $\mathbf{q} = \begin{matrix}-2\\3\end{matrix}$ and $\mathbf{r} = \begin{matrix}3\\-2\end{matrix}$, find q – r.
    A. $\begin{matrix}-5\\5\end{matrix}$
    B. $\begin{matrix}5\\5\end{matrix}$
    C. $\begin{matrix}-1\\1\end{matrix}$
    D. $\begin{matrix}1\\1\end{matrix}$

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